

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 06: ERP/CRM/SCM Integration in Commerce

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Session plan (90 minutes)

- 1 Why ERP/CRM/SCM integration is central to commerce
- 2 The enterprise systems perspective
- 3 Commerce versus enterprise ownership
- 4 Systems of record for key entities
- 5 Master data management (MDM)
- 6 Managerial implications

Learning objectives

By the end of this lecture, you should be able to:

- Explain how commerce platforms integrate with ERP, CRM, and supply-chain systems across operational workflows.
- Identify systems of record for key entities such as customer, product, price list, order, invoice, and shipment.
- Analyze master data management, process ownership, and consistency risks in hybrid commerce architectures.
- Design integration flows for order-to-cash, returns, and B2B pricing using Odoo as a reference ERP environment.

Why ERP/CRM/SCM integration is central to commerce

- orders that appear confirmed online but fail in back-office processing,
- inventory figures that differ between storefront and warehouse,
- duplicate customer records across channels,
- B2B price lists that drift between sales and commerce systems,
- returns and refunds that create reconciliation problems in finance.

The enterprise systems perspective

- **ERP: Sales:** quotations, sales orders, customer terms, sales workflows,; **Inventory:** stock moves, reservations, warehouse operations, replenishment,
- **CRM:** CRM systems focus on relationship history, account management, sales interactions, lead qualification, and customer-service context.
- **SCM and warehouse systems:** Supply-chain and warehouse platforms manage sourcing, supplier coordination, inbound logistics, warehouse execution, and delivery orchestration.

Commerce versus enterprise ownership

- **Common ownership patterns:** The **commerce layer** often owns web sessions, carts, checkout state, personalization context, and channel-specific merchandising behavior.; The **ERP layer** often owns invoices, accounting entries, stock movements, contractual customer records, and internal fulfillment commitments.
- **Why ownership must be explicit:** If pricing is editable in both storefront and ERP, or if both commerce and CRM create customer records independently, inconsistencies are not accidental.

Systems of record for key entities

- **Customer:** the storefront may own channel account credentials and preferences,; ERP may own bill-to and ship-to parties for financial processing,
- **Product:** ERP may own procurement and internal stock-keeping information,; a catalog or PIM layer may own rich digital merchandising content,
- **Price list:** Price lists are especially sensitive in B2B environments.
- **Order:** The commerce platform often captures the initial order intent, but the enterprise question is whether ERP becomes the authoritative record for downstream operational and financial lifecycle.
- **Invoice and payment reconciliation:** Invoices, ledger effects, and payment reconciliation usually belong in ERP or finance-owned systems.
- **Shipment:** Shipment truth may belong in warehouse or logistics systems even if ERP or storefront mirrors the status.

Master data management (MDM)

- **Why MDM matters:** duplicate customer records created by multiple channels,; product records with inconsistent category or attribute definitions,
- **Practical MDM principles:** Define authoritative identifiers and mapping rules.; Separate creation rights from consumption rights.

Order-to-cash in integrated commerce

- **Typical flow:** customer places an order in the storefront,; payment is authorized or payment terms are validated,
- **Architectural decisions inside the flow:** whether the storefront can confirm an order before ERP acceptance,; whether stock is reserved in real time or reconciled later,
- **Failure modes:** the order is captured online but fails to post in ERP,; payment succeeds but the order export fails,

Inventory, fulfillment, and availability

- **Inventory concepts that should not be conflated:** **On-hand inventory:** physical stock currently available in storage.; **Reserved inventory:** stock allocated to pending commitments.
- **Synchronization strategies:** near-real-time events for stock adjustments,; short-lived availability caches,

- **Typical return flow:** customer initiates a return,; policy eligibility is checked,
- **Architectural risks:** the storefront approves a return that finance policy rejects,; inventory is restocked before inspection rules are applied,

B2B pricing and account structures

- **Price lists and negotiated terms:** customer account,; contract terms,
- **Account hierarchies:** a legal entity,; several branches,

Using Odoo as a reference ERP

- **Illustrative Odoo-aligned mapping:** **Sales:** quotations, orders, commercial terms, B2B account workflows.; **Inventory:** stock movements, reservations, warehouse updates.
- **What should remain vendor-neutral:** system-of-record analysis,; event and API contracts,

Integration risks and mitigation strategies

- **Duplicate sources of truth:** mitigated by explicit ownership and reconciliation.
- **Latency mismatch:** mitigated by choosing appropriate sync versus async boundaries.
- **Identifier mismatch:** mitigated by governed master identifiers and mapping tables.
- **Status divergence:** mitigated by explicit lifecycle models and state-mapping rules.
- **Operational blind spots:** mitigated by observability, error queues, and business-level monitoring.

Managerial implications

- Enterprise integration decisions affect more than system design.
- They shape service reliability, customer trust, financial control, and the ability to scale channels.
- If commerce and ERP are tightly coupled in every user interaction, customer experience becomes hostage to back-office latency.